

A binomial count of meromorphic travelling waves

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Abstract

We study meromorphic travelling waves of $u_t + uu_x + \sum_{j=1}^p b_j \partial_x^{j+1} u = 0$, a constant-coefficient family containing KdV–Burgers, Kuramoto–Sivashinsky and Kawahara. After integration and normalization, the profiles satisfy $P(D)v + v^2/2 = 0$, where $D = d/dz$ and P is monic of degree $p \geq 2$ with complex coefficients.

If p is odd or P contains only even powers, we give a complete algebraic description of all nonconstant globally meromorphic solutions: they are rational, rational in an exponential, or elliptic, with one pole modulo their full period group.

We then count normalized one-pole pairs (P, v) across all monic P , whether or not P is covered by the classification. When $2p - 1$ is prime, there are $\binom{2p-1}{p-1}$ rational-exponential pairs at a fixed exponential rate and $2\binom{2p-2}{p-2}$ elliptic pairs on a fixed generic lattice, all of multiplicity one. Exact computer certificates establish both formulas with multiplicities for $2 \leq p \leq 73$. We conjecture the same counting and multiplicity statements in every order.

Keywords. Meromorphic travelling waves; Laurent series; elliptic functions; polynomial convolution; algebraic multiplicity.

1 Introduction

The generalized Kuramoto–Sivashinsky equation has nonconstant travelling waves that are polynomials in a hyperbolic tangent at exactly four values of its squared dimensionless dispersion parameter. Kuramoto and Tsuzuki found the dispersionless one [37], Kudryashov the other three [24, 25], and Eremenko proved that, together with Kudryashov’s elliptic solution and a single rational one, they exhaust the nonconstant meromorphic travelling waves [18]. Counted as we count below, the tanh list has ten members. This paper shows that ten is $\binom{5}{2}$: the classical list is the order-three case of a binomial enumeration, proved for infinitely many orders and conjectured for all.

We consider the evolution equations

$$u_t + uu_x + \sum_{j=1}^p b_j \partial_x^{j+1} u = 0, \quad b_p \neq 0,$$

which contain KdV–Burgers ($p = 2$), Kuramoto–Sivashinsky ($p = 3$) and Kawahara ($p = 4$), the sixth-order turbulence models of Kudryashov and Migita [30], Simbawa et al. [42], and the quadratic long-wave reductions of Kudryashov et al. [32]. A travelling wave $u = w(x - ct)$ satisfies, after one integration, $\sum_j b_j w^{(j)} - cw + w^2/2 + K = 0$. Shifting w by a constant equilibrium and normalizing the leading coefficient gives

$$E_P(v) := P(D)v + \frac{1}{2}v^2 = 0, \quad P(\rho) = \rho^p + \sum_{j=0}^{p-1} a_j \rho^j, \quad p \geq 2, \quad (1.1)$$

with $D = d/dz$ and complex coefficients. The speed, the integration constant and the equilibrium are absorbed into a_0 . Throughout, p is the order of (1.1), one less than the order of the evolution equation.

Completeness. Eremenko's theorem states that an autonomous algebraic differential equation with a single monomial of top degree and finitely many Laurent expansions at a pole has only three kinds of nonconstant meromorphic solutions: rational, rational in an exponential, and elliptic [18]. Theorem 2.1 verifies these hypotheses directly for (1.1) when p is odd, and applies the theorem to a fixed-energy equation when P is an even polynomial. Every nonconstant meromorphic solution is then of one of the three kinds and has a single pole modulo its full period group, and for fixed P there is at most one nonconstant solution up to translation when p is odd and at most one on each energy level when P is even. Section 3 writes each kind as a finite polynomial system, following the principal-part construction of Demina and Kudryashov [15, 16]. These systems therefore give an exhaustive representation of the nonconstant meromorphic travelling waves in the covered cases. The matching systems are valid for every monic P , and the counts below range over the full monic family; parity enters only through exhaustion.

The count. Fix the exponential rate to one and place the pole at $i\pi$. Every solution of (1.1) that is rational in e^z , has one pole modulo $2\pi i\mathbb{Z}$ and vanishes as $e^z \rightarrow 0$ is of the form $v = s_p A(D)Q$, where $Q = 1/(1 + e^{-z})$, s_p is a fixed scale and A is a monic polynomial of degree $p - 1$. Let P range over all monic polynomials of degree p and count the pairs (P, v) for which v solves (1.1). Likewise fix a nonsingular Weierstrass lattice and count the pairs (P, v) with v elliptic on that lattice and its sole pole at zero. We prove

$$\begin{aligned} N_p &= \binom{2p-1}{p-1} \quad \text{rational-exponential pairs,} \\ M_p &= 2 \binom{2p-2}{p-2} \quad \text{elliptic pairs on a generic lattice.} \end{aligned}$$

Whenever $2p - 1$ is prime, both formulas hold and every pair is simple, that is, nondegenerate; in the elliptic case this is for a generic lattice (Theorems 4.4 and 4.7). For every $2 \leq p \leq 73$, the formulas hold as counts with algebraic multiplicity (Section 4). At $p = 5$, where $2p - 1 = 9$ is the first composite index, a separate exact computation shows that all 126 rational-exponential pairs are simple. We conjecture the same counting and simplicity statements in every order (Conjecture 6.1). The count is over operators, not over the waves of a single equation.

For rational-exponential profiles, the pair (P, v_A) solves (1.1) if and only if A divides its own discrete self-convolution $S_A(n) = \sum_{j=1}^{n-1} A(j)A(n-j)$, and then $P = s_p S_A/(2A)$ (Proposition 4.1). When $\ell = 2p - 1$ is prime, $\ell S_A \equiv \pm(\rho^\ell - \rho) \pmod{\ell}$, so the condition reduces to $A \mid \rho^\ell - \rho$ over $\mathbb{F}_\ell$. The monic divisors of degree $p - 1$ are indexed by the $(p - 1)$ -element subsets of $\mathbb{F}_\ell$. There are $\binom{\ell}{p-1}$ of them, and each lifts by Hensel's lemma to a simple ℓ -adic solution. A weighted Bézout bound shows that the lifts account for the whole solution scheme, and finiteness over $\mathbb{Q}$ transfers the count to $\mathbb{C}$. For the elliptic pairs, the same weighted Bézout argument gives the count with multiplicity on every nonsingular lattice, once a cusp degeneration excludes solutions at infinity; degenerating the lattice to a node instead, where the elliptic problem becomes the equal-endpoint case of the exponential one with two choices of background, proves that the pairs are simple on a generic lattice. Two ingredients resist a uniform proof. One is a rigidity statement for the continuous convolution $\int_0^p A(t)A(\rho - t)dt$, needed to exclude solutions at infinity; it is proved

for prime and prime-power indices, certified by exact arithmetic through degree 72, and would give both formulas with multiplicity in every order. The other is simplicity of every rational-exponential pair, proved for prime indices and, by exact computation, at $p = 5$, and open in general (Conjecture 6.1).

Relation to earlier work. The completeness theorem is a uniform version of results that exist in pieces. For odd p the indicial polynomial has no positive integer zero, the Laurent expansion at a pole is unique, and Eremenko’s theorem applies directly; this case is contained in Yuan et al. [45, Theorem 1], with related odd-order results in Huang et al. [21], Yuan et al. [46] and a fifth-order instance in Dang [10]. For even p the index $3p$ is resonant, and the conclusion need not hold for mixed operators: the Fisher equation has an elliptic-exponential family of solutions [1, 5, 28] whose generic members are meromorphic and of none of the three kinds. The familiar Ablowitz–Zeppetella front is a rational-exponential degeneration of this family. When P is an even polynomial the equation has a first integral, the energy, and fixing its value removes the resonant freedom. This is the device of Eremenko et al. [19, Lemmas 3–4] for $P = D^p + a_0$, stated as a general step by Conte et al. [6, 7] and applied to the Kawahara equation [14, 44], the generalized Bretherton equation [15, 36] and the seventh-order KdV equation [9]. Theorem 2.1 extends it uniformly to every even P , intermediate derivatives included. Equations with two dominant monomials, or without the finiteness property, are treated by other methods in Demina [12, 13]. The general enumeration has no precedent that we know of; its special cases at $p = 3$ and in the even Kawahara family are classical, as the examples make explicit.

Examples. The rational-exponential and generic elliptic counts are, respectively, 3 and 2 for KdV–Burgers, 10 and 8 for Kuramoto–Sivashinsky, and 3 and 6 for the Kawahara family, which restricts P to even polynomials, against 35 and 30 for the unrestricted order-four family. The ten Kuramoto–Sivashinsky pairs are the classical list tabulated by Kudryashov and Zargaryan [31] and Conte and Musette [4]; the elliptic solution and its parameter condition are due to Kudryashov [25], with necessity proved by Hone [20] and Eremenko [18]; the three Kawahara values are those of Demina and Kudryashov [14]. The Kuramoto–Sivashinsky example also exhibits the two exceptional values $j = 0$ and $j = -300$ of the lattice invariant at which elliptic pairs collide. All counts are over $\mathbb{C}$; the examples identify the real profiles, and stability is not discussed. Section 6 collects the open questions, and Appendix A describes the exact verification and the ancillary material.

2 Pole structure and completeness

This section proves Theorem 2.1 below: for odd p , and for even polynomials P , every nonconstant meromorphic solution of (1.1) is rational, rational in an exponential, or elliptic, and its poles form a single orbit under its periods. The proof has three steps. We first compute the Laurent expansion at a pole and show that for odd p it is unique. For even P it is not: one coefficient is left free, and we show that the energy, a first integral of (1.1), fixes it, so that Eremenko’s theorem applies on each energy level. Finally we show that any two poles of one solution differ by a period.

Throughout, a meromorphic function is single-valued and meromorphic on the whole complex plane. Let $\mathcal{W}$ consist of rational functions of z , rational functions of $e^{\kappa z}$ with $\kappa \neq 0$, and elliptic functions. The period group of a nonconstant meromorphic function is its *full* group of periods:

trivial for a rational function, infinite cyclic for a rational function of $e^{\kappa z}$, where we choose κ so that $2\pi i/\kappa$ generates it, and the period lattice for an elliptic function. For the general analytic background, see Eremenko [17].

The theorem collects the analytic input to the enumeration. Its odd-order case is contained in Yuan et al. [45, Theorem 1]. For the even-order case, the first-integral mechanism is that of Eremenko et al. [19, Lemmas 3–4]; here it is written uniformly for every even polynomial P . We give the proofs in full because Sections 3 and 4 use the Laurent recurrence and the energy dependence in explicit form.

Theorem 2.1 (Completeness in the covered operator classes). *Assume either that p is odd or that P is an even polynomial. Every nonconstant meromorphic solution of (1.1) belongs to $\mathcal{W}$. Every pole has order p , and the pole set is one coset of the full period group. Consequently there is one pole in the rational case, one pole modulo the primitive period in the simply periodic case, and one pole on the full elliptic period torus.*

For fixed P of odd degree, there is at most one nonconstant solution with its pole at zero. For even P , the same statement holds after fixing the conserved energy. The only entire solutions are $v = 0$ and $v = -2a_0$.

An even polynomial is one in which every odd power of ρ has coefficient zero, so that $P(D)$ contains only even derivatives. The hypothesis does not cover an arbitrary operator of even degree, and it cannot: Section 6 gives a mixed second-order operator with a meromorphic solution outside $\mathcal{W}$.

2.1 Pole order and the Laurent recurrence

Write a pole expansion at zero as

$$v(z) = \sum_{n \geq 0} c_n z^{n-p}. \quad (2.1)$$

The pole order is forced by dominant balance. A pole of order r makes $P(D)v$ of order $r + p$ and v^2 of order $2r$; the two can cancel only if $r = p$. Comparing the leading coefficients gives

$$c_0 = c_* := -2(-1)^p R_p, \quad R_p = \frac{(2p-1)!}{(p-1)!}. \quad (2.2)$$

For the remaining coefficients, set

$$I_p(n) = \prod_{j=p}^{2p-1} (n-j) + c_*. \quad (2.3)$$

The product is the factor produced by D^p acting on z^{n-p} , and c_* is the factor produced by the cross term of $c_n z^{n-p}$ with the leading pole in $v^2/2$. With $c_n = 0$ for $n < 0$, comparison of the coefficients of z^{n-2p} in (1.1) gives

$$I_p(n)c_n = - \sum_{j=0}^{p-1} a_j (n-2p+j) \underline{j} c_{n-p+j} - \frac{1}{2} \sum_{k=1}^{n-1} c_k c_{n-k}, \quad n \geq 1. \quad (2.4)$$

The underline denotes a falling factorial. The right side contains only coefficients with index less than n , so (2.4) determines c_n whenever $I_p(n) \neq 0$. The positive integer zeros of I_p are the

resonances of the recurrence: at such an index the recurrence leaves c_n undetermined if its right side vanishes, and has no solution otherwise.

If p is odd, then $c_* = 2R_p$, and there is no positive integer resonance. For integers $1 \leq n < p$, the product in (2.3) is negative with absolute value less than R_p . For $p \leq n \leq 2p - 1$ it is zero, and for $n \geq 2p$ it is positive. Hence $I_p(n) \neq 0$ for every positive integer n , and there is exactly one formal Laurent expansion at the marked pole.

To compare directly with Yuan et al. [45], put $w = -v/2$. Then $w^{(p)} + \sum_{j < p} a_j w^{(j)} = w^2$. Their pole order is p , their number of principal parts is one, and each lower linear monomial has weighted pole order $p + j < 2p$. Their degree and dominant-term hypotheses therefore hold. We nevertheless apply Eremenko's theorem directly, since the same theorem will be applied to an energy level in the even case.

Theorem 2 of Eremenko [18] states that an autonomous polynomial differential equation with finitely many possible formal pole expansions and a single monomial of highest total degree has all its meromorphic solutions in $\mathcal{W}$. In (1.1) the only monomial of degree two is $v^2/2$, and for odd p we have just seen that there is exactly one formal pole expansion. Both hypotheses hold at every odd order, for every coefficient vector a .

2.2 The energy argument for an even operator

Let $p = 2m$ and

$$P(D) = D^{2m} + \sum_{j=0}^{m-1} b_j D^{2j}. \quad (2.5)$$

Now $c_* = -2R_p$, and there is exactly one positive integer resonance. Indeed the product in (2.3) is at most R_p for $0 \leq n < p$, is zero for $p \leq n \leq 2p - 1$, and is strictly increasing for $n \geq 2p$; its value at $3p$ is $2R_p = -c_*$. The only positive integer zero of I_p is therefore $n = 3p$, and the possible free Laurent coefficient is

$$h = c_{3p} = [z^{2p}]v.$$

If the recurrence is compatible at $n = 3p$, the equation has a one-parameter family of formal expansions at the pole, one for each value of h , and Eremenko's theorem does not apply to (1.1) itself. The remedy is a first integral whose value determines h .

For $j \geq 1$, define the quadratic differential expression

$$\mathcal{B}_j(v) = \sum_{\ell=0}^{j-2} (-1)^\ell v^{(\ell+1)} v^{(2j-1-\ell)} + \frac{(-1)^{j-1}}{2} (v^{(j)})^2, \quad (2.6)$$

where the sum is empty when $j = 1$. Termwise differentiation gives $D\mathcal{B}_j = v'v^{(2j)}$: every even derivative, multiplied by v' , is an exact derivative. Since P contains only even derivatives, it follows that

$$\mathcal{H}_P(v) = \mathcal{B}_m(v) + \sum_{j=1}^{m-1} b_j \mathcal{B}_j(v) + \frac{b_0}{2} v^2 + \frac{1}{6} v^3 \quad (2.7)$$

satisfies

$$D\mathcal{H}_P = v'E_P(v). \quad (2.8)$$

In particular $\mathcal{H}_P$, the energy, is constant on each solution. The case with no intermediate derivatives is the first integral used by Eremenko et al. [19]. The familiar Kawahara first integral is the case $m = 2$ [14, 44]; the corresponding order-six construction appears in Dang [9].

The energy identity has two consequences for the Laurent expansion. The first is that the resonance is always compatible. To see this, construct (2.1) through index $3p - 1$. The first possible residual of (1.1) is $\Omega_p z^p$. Multiplication by v' gives residue $-pc_*\Omega_p$. By (2.8) this is the residue of a derivative of a Laurent series, and is therefore zero. Hence $\Omega_p = 0$, and h may be chosen arbitrarily.

The second consequence is that the energy determines h . On the full formal solution the constant energy has the form

$$\mathcal{H}_P = \Lambda_p h + \eta_p(P), \quad \Lambda_p = -pc_* I'_p(3p) \neq 0. \quad (2.9)$$

To find the coefficient, linearize the highest-weight part of (2.8) about $c_* z^{-p}$, with perturbation z^{n-p} . If the corresponding energy term is $L(n)z^{n-3p}$, then

$$(n - 3p)L(n) = -pc_* I_p(n).$$

Taking $n \rightarrow 3p$ gives the coefficient in (2.9). Lower derivative terms have different powers of z and cannot contribute to this coefficient. Moreover,

$$I'_p(3p) = 2R_p \sum_{j=p+1}^{2p} \frac{1}{j} > 0.$$

Fixing the energy thus fixes h , and all subsequent Laurent coefficients follow from (2.4). This is the resonance-removal step in Eremenko et al. [19, Lemma 4], with an explicit coefficient for the present operator family; see also Conte et al. [6, Section 3.2] and Conte et al. [7, Section 5].

We can now apply Eremenko's theorem, not to (1.1) but to the level equation $\mathcal{H}_P(v) = H$ with P and H fixed. This equation is autonomous and polynomial, and its unique monomial of highest degree is $v^3/6$. It has exactly one formal pole expansion: differentiating it and dividing by the nonzero Laurent series v' returns $E_P(v) = 0$, so every expansion belongs to the family above, and (2.9) selects the value of h . Every meromorphic solution of (1.1) lies on one energy level and therefore belongs to $\mathcal{W}$.

2.3 The period group and the entire solutions

It remains to show that the poles of one solution form a single orbit under its periods. For either covered case, take two poles z_1, z_2 of the same solution. The functions $v(z + z_1)$ and $v(z + z_2)$ satisfy the same equation and, in the even case, have the same energy. Their Laurent series at zero coincide, so the identity theorem gives

$$v(z + z_1) = v(z + z_2).$$

Every difference of poles is a period. Conversely, every period translates a pole to a pole. The period group is discrete: a sequence of nonzero periods tending to zero would force $v' \equiv 0$. The pole set is therefore one coset of that group. The same argument proves the uniqueness statements of the theorem: two nonconstant solutions with the same operator, the same energy in the even case, and a pole at zero have the same Laurent series there.

Finally, an entire solution in $\mathcal{W}$ must be constant. In the rational case it is a polynomial, and the highest degree of its square cannot be cancelled in (1.1) unless the polynomial is constant. In the rational-exponential case it is a Laurent polynomial in $t = e^{\kappa z}$; the operator $P(\kappa t \partial_t)$ preserves each exponent, whereas squaring doubles an extreme nonzero exponent. An entire

elliptic function is constant by compactness. The constants solve $a_0v + v^2/2 = 0$, which gives $v = 0$ and $v = -2a_0$. This completes the proof of Theorem 2.1.

The finite systems of the next section are derived from the one-pole structure alone. The analytic facts of this section are used only to show that, in the covered cases, those systems miss no nonconstant meromorphic solution. The systems themselves never require the convergence of a formal series.

3 Three finite algebraic systems

Theorem 2.1 says that, in the covered cases, a nonconstant meromorphic solution is one of three kinds and has one pole modulo its periods. Each kind is a finite-dimensional family of functions: a polynomial in $1/z$, a polynomial in a fixed simply periodic function with one pole, or a constant plus a linear combination of $\wp$ and its derivatives. Substituting the general member of each family into (1.1) therefore gives finitely many polynomial equations in its coefficients. This section writes down the three systems, fixes their normalizations, and proves that they are exactly equivalent to the one-pole solutions (Theorem 3.1). They are instances of the principal-part construction of Demina and Kudryashov [15, 16].

The systems all have the same shape. Let V be a polynomial expression in some variables, and let δ be a differentiation rule for those variables, so that the polynomials in them form a differential algebra with derivation δ . Write

$$\mathcal{E}_P(V; \delta) = \delta^p V + \sum_{j=0}^{p-1} a_j \delta^j V + \frac{1}{2} V^2.$$

Each system below is the polynomial identity $\mathcal{E}_P(V; \delta) = 0$ for an appropriate V and δ . A polynomial identity means the vanishing of every coefficient.

3.1 Rational solutions

A rational solution whose only pole is at zero, of order p , and which has no nonconstant polynomial part at infinity is a polynomial of degree p in $T = 1/z$. Since $DT = -T^2$, the equation becomes

$$V_R(T) = b + \sum_{k=1}^p r_k T^k, \quad r_p = c_*, \quad \mathcal{E}_P(V_R; -T^2 \partial_T) = 0. \quad (3.1)$$

The left side of the identity is a polynomial of degree at most $2p$ in T , so this is a finite system in $b, r_1, \dots, r_{p-1}$. No rational solution has a nonconstant polynomial part at infinity: such a part would have its degree doubled by v^2 , while the linear differential expression cannot reach that degree. Consequently (3.1) is the full rational one-pole system.

3.2 Rational functions of an exponential

For $q = \kappa^2 \neq 0$, define

$$T_q(z) = \frac{\kappa}{2} \coth \frac{\kappa z}{2}, \quad DT_q = \frac{q}{4} - T_q^2. \quad (3.2)$$

The expression is independent of the choice of sign of κ . It has period $2\pi i/\kappa$ and one pole per period, at zero, with residue one; it plays the role that $1/z$ plays in the rational case. The finite

system is

$$V_S(T) = \sum_{k=0}^p u_k T^k, \quad u_p = c_*, \quad q \neq 0, \quad \mathcal{E}_P \left(V_S; \left(\frac{q}{4} - T^2 \right) \partial_T \right) = 0. \quad (3.3)$$

To see that every one-pole solution rational in an exponential has this form, write $v = R(t)$, $t = e^{\kappa z}$. The rational function R has finite limits at both $t = 0$ and $t = \infty$. If it had a pole at either end, an extreme Laurent monomial would be doubled by R^2 and could not cancel against $P(\kappa t \partial_t)R$. After a translation, its only finite pole is at $t = 1$, of order p . Such a function is a degree- p polynomial in T_q . The logistic coordinate

$$Q = \frac{1}{1 + e^{-z}}, \quad DQ = Q(1 - Q),$$

is an equivalent coordinate after fixing the rate and translating the pole; $Q(z + i\pi) = T_1(z) + 1/2$. Polynomial logistic representations and their use in constructing travelling waves are developed by Kudryashov [29, 33, 34].

The rational system is the $q = 0$ specialization of (3.3), with $T_0 = 1/z$. We keep $q = 0$ and $q \neq 0$ distinct so that this degeneration is never counted twice.

3.3 Elliptic solutions

Use the Weierstrass algebra

$$Y^2 = 4X^3 - g_2X - g_3, \quad \mathcal{D}X = Y, \quad \mathcal{D}Y = 6X^2 - \frac{g_2}{2}, \quad \Delta = g_2^3 - 27g_3^2 \neq 0. \quad (3.4)$$

Its analytic realization is $X = \wp(z; g_2, g_3)$, $Y = \wp'(z; g_2, g_3)$; see NIST [41, Chapter 23]. Set

$$V_E = b + \sum_{j=0}^{p-2} t_j \mathcal{D}^j X, \quad t_{p-2} = -s_p, \quad s_p = \frac{2(2p-1)!}{((p-1)!)^2}, \quad \mathcal{E}_P(V_E; \mathcal{D}) = 0. \quad (3.5)$$

Using (3.4), every polynomial in X and Y reduces to the normal form $F(X) + YG(X)$. The system consists of the coefficients of F and G set equal to zero; all of them are polynomials with rational coefficients in the displayed unknowns.

The representation is the classical one. An elliptic function with one pole has zero residue there. The functions $1, \wp, \wp', \dots, \wp^{(p-2)}$ have distinct principal parts and span the space of elliptic functions whose only pole is at zero, of order at most p . Subtracting the indicated principal part from a one-pole solution leaves an entire elliptic function, hence a constant. The leading coefficient in (3.5) follows from

$$D^{p-2}\wp = (-1)^{p-2}(p-1)!z^{-p} + O(1).$$

The nonsingularity condition $\Delta \neq 0$ is retained throughout; the singular cubics correspond to the rational and simply periodic systems, which we have already written down.

Theorem 3.1 (Finite matching and exhaustion). *For every $p \geq 2$, the systems (3.1), (3.3), and (3.5) are necessary and sufficient for the respective one-pole solutions of (1.1), with the pole at zero and with the stated normalizations. For the operator classes of Theorem 2.1, these systems, together with the constant solutions, exhaust all globally meromorphic solutions.*

The systems describe exceptional parameter values and continuous families as well as isolated solutions. Their sufficiency follows from substitution, without any genericity assumption.

Proof. Necessity is the principal-part argument given for each kind above. Conversely, a solution of a coefficient system gives a globally meromorphic function by its displayed realization, and the chain rule gives $E_P(v) = 0$. The leading coefficient $c_* \neq 0$ ensures a nonconstant function. In the elliptic case every nonsingular pair (g_2, g_3) has a Weierstrass realization, and the normal form $F(X) + YG(X)$ is unique. The exhaustion statement is Theorem 2.1. $\square$

3.4 Eliminating the profile coefficients

The systems above treat the operator and the profile coefficients as unknowns together, which is the form the enumeration in Section 4 requires. For a fixed operator they can be reduced further, because the Laurent recurrence already determines the principal part of any solution. Compute

$$r_k = c_{p-k} \quad (1 \leq k \leq p), \quad s = c_p$$

from (2.4). These coefficients occur before the even-order resonance and are determined by P at either parity. In the simply periodic algebra define

$$S_k(T; q) = \frac{(-1)^{k-1}}{(k-1)!} \left[\left(\frac{q}{4} - T^2 \right) \partial_T \right]^{k-1} T.$$

It has principal part z^{-k} . Its regular Laurent constant is

$$d_k(q) = \begin{cases} 0, & k \text{ odd}, \\ -B_k q^{k/2}/k!, & k \text{ even}, \end{cases}$$

where B_k denotes a Bernoulli number. Thus the unique candidate with the required principal part and constant is

$$V_0(T; P, q) = s + \sum_{k=1}^p r_k (S_k(T; q) - d_k(q)). \quad (3.6)$$

Substituting it into (1.1) leaves polynomial equations in the single unknown q .

For the elliptic candidate, first require $r_1 = 0$, since an elliptic function with one pole has no residue there, and put

$$\tilde{V}_E = s + \sum_{k=2}^p r_k \left(\frac{(-1)^{k-2}}{(k-1)!} \mathcal{D}^{k-2} X - e_k(g_2, g_3) \right), \quad (3.7)$$

where e_k is the regular Laurent constant of the corresponding derivative of $\wp$. It is a polynomial in g_2, g_3 , computed from (3.4). The remaining equations involve only g_2, g_3 , with $\Delta \neq 0$. This is the coefficient-comparison construction used in Demina and Kudryashov [11, 15].

For even P the equation $E_P(v) = 0$ does not see the energy: a candidate may solve it on any energy level. To classify a fixed level, compute h from (2.9) for the prescribed energy and append

$$[z^{2p}]V_0(T_q; P, q) = h \quad \text{or} \quad [z^{2p}]\tilde{V}_E(\wp, \wp'; P, g_2, g_3) = h. \quad (3.8)$$

Both are polynomial equations obtained from finite Laurent expansions. With this equation included, the reduced systems classify a fixed energy level: at fixed P of odd degree, or fixed (P, h) in the even case, there is at most one nonconstant solution with its pole at zero.

When all prescribed data are exact algebraic numbers, that is, the coefficients of P and, in the even case, the energy, these are finitely many polynomial equations in one or two unknowns.

Polynomial gcds in the one-variable case and elimination in the multivariable case decide whether a solution exists. A nonvanishing condition such as $\Delta \neq 0$ is imposed by adjoining $u\Delta - 1 = 0$. Elimination with case distinctions describes instead the operators, or the marked data (P, h) , that admit each kind of solution, as a finite union of systems of polynomial equations and nonvanishing conditions [8].

4 Enumeration at fixed scale

In this section P ranges over *all* monic polynomials of degree p , including mixed operators of even degree, and we count the pairs (P, v) in which v is a one-pole solution of (1.1) of a given kind: rational in an exponential at a fixed rate here, elliptic on a fixed lattice in Section 4.4. Nothing in the enumeration requires p to be odd or P to be an even polynomial. Those hypotheses were needed to show that the one-pole solutions are all the meromorphic solutions; the matching equations and their solution counts make sense without them. The counts are for the full monic family and change for a restricted family such as the even polynomials P ; Section 5.3 gives an example.

For the rational-exponential kind, fix the rate to one, the pole to $i\pi$ modulo $2\pi i\mathbb{Z}$, and the value at $e^z = 0$ to zero. Reflection is not identified: distinct normalized pairs related by $z \mapsto -z$ are counted separately. The functions $Q, DQ, \dots, D^{p-1}Q$ span the degree- p polynomials in Q with zero constant term, since

$$D^j Q = (-1)^j j! Q^{j+1} + \text{lower powers of } Q.$$

The leading pole coefficient (2.2) therefore gives the unique form

$$v_A = s_p A(D)Q, \quad A(\rho) = \rho^d + \alpha_1 \rho^{d-1} + \dots + \alpha_d, \quad d = p - 1. \quad (4.1)$$

The scale s_p was defined in (3.5). This representation is the standard singular-part operator form: $Q = 1/2 + D \log \cosh(z/2)$, so (4.1) is a constant plus a linear differential operator applied to $\log \cosh(z/2)$. Such forms, including the KS case, are discussed by Conte and Musette [3]. We use the coefficients $\alpha_1, \dots, \alpha_d$ of that operator as the coordinates for the enumeration.

The argument runs as follows. The profile v_A solves (1.1) for some monic P exactly when A divides a polynomial built from A by a discrete convolution (Proposition 4.1). The remainder of that division gives d polynomial equations in the d coordinates. We count their solutions with multiplicity by Bézout's theorem, using a weighted degree adapted to the equations (Theorem 4.3). When $2p - 1$ is prime, reduction modulo $2p - 1$ identifies the solutions one by one (Theorem 4.4).

4.1 Discrete convolution and the matching equations

For polynomials F, G , define $F \star G$ by polynomial continuation of

$$(F \star G)(n) = \sum_{j=1}^{n-1} F(j)G(n-j), \quad n \geq 1.$$

The continuation exists by the power-sum formulas: for fixed F, G the right side is a polynomial in n . Write

$$S_A = A \star A.$$

For monic A of degree d ,

$$\deg S_A = 2d + 1, \quad [\rho^{2d+1}]S_A = \beta_d := \frac{(d!)^2}{(2d+1)!}. \quad (4.2)$$

The leading coefficient is the beta integral $\int_0^1 t^d(1-t)^d dt$.

Proposition 4.1 (Polynomial divisibility). *The profile (4.1) solves (1.1) for some monic P if and only if*

$$A \mid S_A. \quad (4.3)$$

The operator is unique and is

$$P_A = \frac{s_p S_A}{2A}. \quad (4.4)$$

Proof. In the half-plane $\operatorname{Re} z < 0$, the logistic function is a geometric series in e^z , and applying $A(D)$ multiplies the n th term by $A(n)$:

$$A(D)Q = \sum_{n \geq 1} (-1)^{n-1} A(n) e^{nz}.$$

Squaring gives the rational-function identity

$$(A(D)Q)^2 = -S_A(D)Q. \quad (4.5)$$

The map $B \mapsto B(D)Q$ is injective: its vanishing would give $B(n) = 0$ for every positive integer n . Consequently (1.1) is equivalent to $2P_A A = s_p S_A$, which has a polynomial solution P_A exactly when A divides S_A . Equation (4.2) and $s_p = 2/\beta_d$ show that P_A is monic of degree p . $\square$

The operator determines the profile. This holds for every p , without the parity hypotheses of Theorem 2.1, because it only uses the leading coefficient of a convolution.

Proposition 4.2 (One profile per operator). *Let A, B be monic polynomials of degree $p-1$ satisfying $A \mid S_A$ and $B \mid S_B$. If $P_A = P_B$, then $A = B$. Consequently a monic operator of degree p admits at most one normalized rational-exponential one-pole profile. Any count of distinct pairs with this normalization is therefore also a count of distinct operators.*

Proof. Suppose $P_A = P_B = P$, and put $h = A - B$. If $h \neq 0$, its degree e is less than $d = p-1$; write $f \neq 0$ for its leading coefficient. By (4.4) and the bilinearity and symmetry of the convolution,

$$S_A - S_B = h \star (A + B) = \frac{2}{s_p} P h.$$

The power-sum calculation in (4.2) gives the leading coefficient $2f \int_0^1 t^e(1-t)^d dt$ on the left. Since P is monic, the right side has leading coefficient $(2/s_p)f = \beta_d f$. Cancelling f would therefore give

$$2 \int_0^1 t^e(1-t)^d dt = \int_0^1 t^d(1-t)^d dt.$$

This is impossible: $e < d$ implies $t^e > t^d$ for $0 < t < 1$, so the left side exceeds twice the right side. Thus $h = 0$. $\square$

For odd p , Theorem 2.1 gives uniqueness among all nonconstant meromorphic solutions up to translation; for even P , it does so on a fixed energy level. Proposition 4.2 fixes the rational-exponential normalization and holds for every monic operator.

Divide S_A by the monic polynomial A :

$$S_A = AC_A + R_A, \quad R_A = \sum_{r=0}^{d-1} R_r(\alpha_1, \dots, \alpha_d) \rho^r. \quad (4.6)$$

The matching equations are $R_0 = \dots = R_{d-1} = 0$: d polynomial equations in the d unknowns $\alpha_1, \dots, \alpha_d$. Their solutions are in bijection with the normalized pairs (P_A, v_A) . We write $\mathcal{X}_p$ for the solution set together with the equations that define it; in algebraic language, $\mathcal{X}_p$ is the affine scheme defined by the R_r .

Three notions from algebra are used to count. A solution is *simple* if the Jacobian matrix of $(R_0, \dots, R_{d-1})$ with respect to $(\alpha_1, \dots, \alpha_d)$ is invertible there; equivalently, the equations admit no nonzero infinitesimal deformation of the solution. Every isolated solution has a *multiplicity*, the dimension over $\mathbb{C}$ of the local quotient algebra of the equations at that point. For a square system such as this one it may be pictured as the number of simple solutions into which the point splits under a generic small perturbation of the equations; it equals one exactly for simple solutions. When $\mathcal{X}_p$ is finite, its *length* is the sum of the multiplicities of its points, and $\mathcal{X}_p$ is *reduced* when every point is simple, so that the length is the number of points. An isolated solution of multiplicity greater than one is still the only solution near itself; local uniqueness is therefore weaker than simplicity.

4.2 Weighted degree and the role of continuous convolution

Bézout's theorem bounds the sum of the multiplicities of the isolated solutions of d polynomial equations in d unknowns by the product of their degrees. Equality holds when the system has no solutions at infinity, that is, when the highest-degree parts of the equations have no common zero other than the origin. The ordinary degrees of the matching equations give a poor bound. The right measure of their size is a weighted degree that reflects how the α_j scale. Give α_j weight j , and ρ weight one. The remainder equations have highest weighted degrees

$$2d + 1 - r \quad (0 \leq r < d), \quad \text{namely } d + 2, \dots, 2d + 1. \quad (4.7)$$

Their highest weighted parts are the coefficients of $\text{rem}_A H_A$, where

$$H_A(\rho) = \int_0^\rho A(t) A(\rho - t) dt. \quad (4.8)$$

Indeed, if $A(\rho) = \sum a_i \rho^i$, then

$$H_A(\rho) = \sum_{i,j=0}^d \frac{i!j!}{(i+j+1)!} a_i a_j \rho^{i+j+1}.$$

These are precisely the highest weighted terms of the power sums defining S_A , and monic division preserves the highest weighted part. Indeed, A is weighted-homogeneous of weight d , and each step of long division subtracts a multiple of A of the same weight as the term being removed. In words, the discrete convolution S_A is a lower-order perturbation of the continuous convolution H_A , and the behaviour of the matching equations at infinity is governed by the continuous problem $A \mid H_A$.

For this degree, denote by (R_d) the condition

$$A \text{ monic, } \deg A = d, \quad A \mid H_A \implies A(\rho) = \rho^d. \quad (4.9)$$

It says that the continuous problem has only the trivial solution. By the preceding paragraph, it is exactly the condition that the matching equations have no solution at weighted infinity.

Theorem 4.3 (Count with algebraic multiplicities). *For every $p \geq 2$, the sum of multiplicities of the isolated points of $\mathcal{X}_p$ is at most*

$$N_p = \binom{2p-1}{p-1}.$$

If (R_{p-1}) holds, then $\mathcal{X}_p$ is finite and

$$\text{length } \mathcal{X}_p = N_p. \quad (4.10)$$

This equality does not require simplicity.

Proof. Weighted degree is turned into ordinary degree by the finite substitution

$$\alpha_j = y_j^j, \quad 1 \leq j \leq d. \quad (4.11)$$

A monomial of weight w in the α_j becomes a monomial of ordinary degree w in the y_j , so the pulled-back equations have ordinary degrees at most $d+2, \dots, 2d+1$, and their highest-degree parts are the pull-backs of the highest weighted parts. Bézout's inequality for isolated solutions bounds the sum of their multiplicities by $(d+2) \cdots (2d+1)$; see Mondal [39, Chapter VIII]. If (R_d) holds, the leading homogeneous forms have no nonzero common zero. The homogenized system has no point at infinity, and ordinary Bézout gives equality [8].

It remains to return from the y_j to the α_j . A solution in the α_j with all $\alpha_j \neq 0$ has $1 \cdot 2 \cdots d = d!$ preimages in the y_j , one for each choice of the roots $y_j = \alpha_j^{1/j}$, each with the original multiplicity. In general, for each solution the sum of the multiplicities of its preimages is $d!$ times its original multiplicity, including at branch points where some α_j vanish. Thus the substitution multiplies the total length by $d!$. Algebraically, $\mathbb{C}[y_1, \dots, y_d]$ is a free module over $\mathbb{C}[\alpha_1, \dots, \alpha_d]$, with basis

$$y_1^{e_1} \cdots y_d^{e_d}, \quad 0 \leq e_j < j.$$

This remains true after imposing the equations and restricting to an isolated component. Dividing the Bézout count by $d!$ gives

$$\frac{(d+2) \cdots (2d+1)}{d!} = \binom{2d+1}{d}.$$

The no-infinity condition also gives finiteness of the whole affine scheme, completing the proof. $\square$

For $p = 3$, the variable weights are 1, 2 and the equation weights are 4, 5, so the calculation is $(4 \cdot 5)/(1 \cdot 2) = 10$.

4.3 The prime-index theorem

Theorem 4.4 (Prime-index enumeration). *If $\ell = 2p - 1$ is prime, then*

$$\#\mathcal{X}_p(\mathbb{C}) = \text{length } \mathcal{X}_p = \binom{\ell}{p-1},$$

and every point is simple. After a choice of an ℓ -adic embedding of the algebraic coefficients, reduction labels the points by the $(p-1)$ -element subsets $S \subset \mathbb{F}_\ell$:

$$\overline{A}(\rho) = \prod_{s \in S} (\rho - s). \quad (4.12)$$

Proof. Put $d = p - 1$, so $\ell = 2d + 1$, and work over $\mathbb{Z}_{(\ell)}$, the rationals integral at ℓ . The proof has four steps: a congruence for S_A modulo ℓ ; the solution of the reduced equations over $\mathbb{F}_\ell$; the exclusion of solutions at infinity, which proves (R_d) ; and the lifting of the reduced solutions back to characteristic zero.

Step 1: the congruence. Induction using $(D - n)Q^n = -nQ^{n+1}$ gives

$$Q^n = \frac{(-1)^{n-1}}{(n-1)!} \prod_{j=1}^{n-1} (D - j)Q.$$

Write $(A(D)Q)^2 = \sum_{n=2}^{2p} b_n Q^n$. Each b_n is an integer polynomial in the coefficients of A , and $b_{2p} = (d!)^2$. By (4.5),

$$S_A(\rho) = \sum_{n=2}^{2p} \frac{(-1)^n b_n}{(n-1)!} \prod_{j=1}^{n-1} (\rho - j).$$

Only the last denominator, $(2p-1)! = \ell!$, contains ℓ . Consequently ℓS_A is integral and

$$\ell S_A(\rho) \equiv \frac{(d!)^2}{(\ell-1)!} \prod_{j=1}^{\ell} (\rho - j) \equiv (-1)^d (\rho^\ell - \rho) \pmod{\ell}. \quad (4.13)$$

The scalar follows by pairing j with $\ell - j$ in $(\ell-1)!$, and the product is $\rho^\ell - \rho$ because it runs over all residues modulo ℓ .

Step 2: the reduced equations. The scaled remainder equations $\ell R_r = 0$ therefore reduce exactly to $\overline{A} \mid \rho^\ell - \rho$. There are $\binom{\ell}{d}$ monic divisors of degree d , all defined over $\mathbb{F}_\ell$, and these exhaust the solutions over its algebraic closure. Each of them is a simple solution of the reduced equations. Write $\rho^\ell - \rho = \overline{A}C$. For a variation h with $\deg h < d$, the derivative of the remainder map is, up to the nonzero scalar $(-1)^d$,

$$h \longmapsto -\text{rem}_{\overline{A}}(Ch). \quad (4.14)$$

Since $\rho^\ell - \rho$ is squarefree, $\overline{A}$ and C are coprime, so multiplication by C is invertible modulo $\overline{A}$, and the map (4.14) is invertible.

Step 3: no solutions at infinity. We must also prove that characteristic-zero solutions cannot escape to infinity, that is, that (R_d) holds. Taking the highest weighted part of (4.13) gives

$$\ell H_A(\rho) \equiv (-1)^d \rho^\ell \pmod{\ell}. \quad (4.15)$$

Thus the reduced leading remainder equations have only $\overline{A} = \rho^d$. After (4.11), they are homogeneous equations with no nonzero common zero over $\overline{\mathbb{F}}_\ell$. The same is true in characteristic zero.

If the equations had a complex projective zero, they would also have one with algebraic coordinates, since their coefficients are rational. Normalization at an ℓ -adic place would then produce a nonzero reduced point. The valuation argument is stated precisely in Appendix A. We have proved (R_d) , so Theorem 4.3 gives total length $\binom{\ell}{d}$.

Step 4: lifting. The multivariable simple-root Hensel lemma lifts each of the $\binom{\ell}{d}$ reduced points to a unique point over $\mathbb{Z}_\ell$; see Conrad [2]. The lifts are distinct and simple. Their number already equals the characteristic-zero total length, so they exhaust the scheme. The equations define a finite algebra over $\mathbb{Q}$, hence every coordinate of a solution is algebraic over $\mathbb{Q}$, and the number of geometric points and their local lengths are the same over $\overline{\mathbb{Q}_\ell}$ and over $\mathbb{C}$. This proves the assertion and the labelling. $\square$

Remark 4.5 (Formal verification). The count and simplicity assertions of Theorem 4.4, and (R_{p-1}) for $2p-1$ prime, have been verified in Lean 4 against the Mathlib library; Appendix A describes the files. The formal proof of the count does not use Theorem 4.3. It shows directly that every complex matching point is integral at any valuation ring of $\mathbb{C}$ whose maximal ideal contains ℓ : the roots of A are scaled by a root of smallest valuation, and ℓS_A evaluated at that root has one unit term, all others lying in the maximal ideal. Reduction of integral points is injective, which gives finiteness and the upper bound $\binom{\ell}{d}$, and the same normalization gives simplicity. For the lower bound, each reduced point is lifted over $\mathbb{Z}_\ell$ by a multivariable Hensel lemma derived from formal smoothness, and the resulting family of distinct $\mathbb{Q}_\ell$ -points is transferred to $\mathbb{C}$ through the Nullstellensatz, without identifying $\mathbb{Q}_\ell$ with a subfield of $\mathbb{C}$. The labelling (4.12) is the injective reduction map of that proof.

The labels (4.12) describe reductions of algebraic coefficients; they are not assertions that the complex roots of A , or the characteristic roots of P_A , are integers. For example, $A = \rho^2 + 1/11$ at $p = 3$ reduces to $(\rho - 2)(\rho - 3)$ modulo five.

The endpoints of v_A are its limits as $z \rightarrow \mp\infty$, namely 0 and $s_p A(0)$. Pairs with $A(0) = 0$ have equal endpoints and are pulses; the others are fronts. The pulses are needed separately, because the elliptic count of Section 4.4 degenerates onto them.

Proposition 4.6 (Equal endpoints). *Assume (R_{p-1}) , and let $\mathcal{Z}_p = \mathcal{X}_p \cap \{A(0) = 0\}$, with its induced scheme structure. Then*

$$\text{length } \mathcal{Z}_p = S_p := \binom{2p-2}{p-2}. \quad (4.16)$$

If $\mathcal{X}_p$ is reduced, these are S_p distinct pairs and $P_A(0) \neq 0$ at each of them. In particular, under the prime-index hypothesis, exactly

$$\binom{2p-2}{p-2}$$

normalized pairs have equal zero endpoints. The remaining $\binom{2p-2}{p-1}$ have unequal endpoints. These are counts over $\mathbb{C}$.

Proof. Polynomial summation gives $S_A(0) = -A(0)^2$, so the constant remainder satisfies the polynomial identity

$$R_0 = -A(0)(A(0) + C_A(0)). \quad (4.17)$$

Set $\alpha_d = A(0) = 0$. The equation $R_0 = 0$ disappears, leaving $d-1$ equations of weights $d+2, \dots, 2d$ in variables of weights $1, \dots, d-1$. Their leading forms have no nonzero common

zero by (R_d) . The weighted calculation in Theorem 4.3 gives

$$\text{length } \mathcal{Z}_p = \frac{(d+2) \cdots (2d)}{(d-1)!} = \binom{2d}{d-1}.$$

For $d = 1$ the system is empty and its length is one.

If $\mathcal{X}_p$ is finite and reduced, its coordinate algebra is a product of copies of $\mathbb{C}$; the same is true of any quotient, in particular that of $\mathcal{Z}_p$, so $\mathcal{Z}_p$ is reduced as well. At a point with $A(0) = C_A(0) = 0$, the differential of (4.17) vanishes. Such a point cannot be simple in the square matching system. Therefore $C_A(0) \neq 0$, and $P_A(0) = s_p C_A(0)/2 \neq 0$, at every point of $\mathcal{Z}_p$ when $\mathcal{X}_p$ is reduced. The prime-index assertions now follow from Theorem 4.4 and $N_p - S_p = \binom{2p-2}{p-1}$. $\square$

Under the labelling (4.12) of Theorem 4.4, a pair is a pulse exactly when its label set contains 0. Indeed $A(0) = 0$ forces 0 to be a root of the reduced polynomial $\bar{A}$, and the two counts agree: the $(p-1)$ -element subsets of $\mathbb{F}_\ell$ containing 0 number $\binom{2p-2}{p-2} = S_p$, and those avoiding 0 number $\binom{2p-2}{p-1} = N_p - S_p$.

4.4 Elliptic pairs on a fixed lattice

We now count the elliptic pairs. Fix a lattice $\Lambda \subset \mathbb{C}$, including its scale, and put $g_i = g_i(\Lambda)$. We place the pole at zero, allow P to vary over all monic polynomials of degree p , and retain the constant term of the profile. Neither reflection nor a lattice automorphism is factored out. By Section 3, every profile has the form

$$V = -s_p \mathcal{D}^{p-2} X + \sum_{j=0}^{p-3} u_j \mathcal{D}^j X + b. \quad (4.18)$$

Its only pole on $\mathbb{C}/\Lambda$ is zero, of order p . In particular Λ is its full period lattice. Let $\mathcal{E}_p(g_2, g_3)$ denote the matching equations obtained by substituting (4.18) in (1.1), in the unknowns $a_0, \dots, a_{p-1}$, $u_0, \dots, u_{p-3}$ and b , together with their solution set. The operator coefficients may equivalently be eliminated first, as in the proof below. Length, simplicity and reducedness are understood as in Section 4.

Theorem 4.7 (Fixed-lattice elliptic enumeration). *Assume (R_{p-1}) . For every fixed nonsingular lattice,*

$$\text{length } \mathcal{E}_p(g_2, g_3) = M_p := 2 \binom{2p-2}{p-2}. \quad (4.19)$$

If, in addition, $\mathcal{X}_p$ is reduced, then $\mathcal{E}_p(g_2, g_3)$ is reduced for (g_2, g_3) in a nonempty Zariski-open subset of $\{g_2^3 - 27g_3^2 \neq 0\}$. In particular, if $2p-1$ is prime, a generic fixed lattice admits exactly M_p normalized elliptic operator-profile pairs, all simple.

Thus the elliptic assertion on an arbitrary lattice is a count with multiplicities. Exceptional lattices, on which distinct pairs merge, do occur and cannot be discarded when counting distinct solutions; Section 5.2 gives explicit collisions at order three.

The proof has five steps. We eliminate the operator, which leaves $p-1$ equations in the $p-1$ profile unknowns. We assign weights to the unknowns, under which the highest weighted parts of the equations are obtained by setting $g_2 = g_3 = 0$. We show that at this cusp the equations have only the trivial common zero; this is where (R_{p-1}) enters, and it means that there are no

solutions at weighted infinity for any g_2, g_3 . Weighted Bézout then gives the length M_p on every lattice. Finally, at a nodal cubic the elliptic problem becomes the equal-endpoint case of the rational-exponential problem, whose solutions are simple by Proposition 4.6 when $\mathcal{X}_p$ is reduced, and simplicity spreads from the nodal parameter value to nearby nonsingular parameters.

Proof. Step 1: eliminating the operator. The coefficient at pole order $2p$ vanishes by the choice of s_p . The coefficients at orders $2p-1, \dots, p$ determine $a_{p-1}, \dots, a_0$ successively. At each step the coefficient of the new a_j is the nonzero rational leading coefficient of $D^j V$. This is a polynomial elimination, valid over any $\mathbb{Q}$ -algebra. The residual has pole order at most $p-1$ and zero residue. It lies in the span of

$$\{1\} \cup \{\mathcal{D}^j X : 0 \leq j \leq p-3\}.$$

Hence $p-1$ equations remain in the $p-1$ profile coefficients u_j, b .

Step 2: weights. Give D, X, g_2, g_3 weights $1, 2, 4, 6$, respectively, and give the profile weight p . Then u_j has weight $p-j-2$, and b has weight p . After the triangular elimination, the residual principal-part coefficients and the residual Laurent constant have weights

$$\begin{array}{l|l} \text{variables} & 1, 2, \dots, p-2, p \\ \text{equations} & p+1, p+2, \dots, 2p-2, 2p. \end{array} \quad (4.20)$$

The Laurent expansions and the eliminations are polynomial in g_2, g_3 , so these statements extend to the singular cubics. For fixed g_2, g_3 , the highest weighted parts of the equations in the profile variables are obtained by setting $g_2 = g_3 = 0$, since g_2 and g_3 carry positive weight.

Step 3: the cusp. At the cusp $g_2 = g_3 = 0$, $X = z^{-2}$ and $Y = -2z^{-3}$, so the profile is a rational function. Write $V = b + f(z)$, where f has its sole pole at zero and vanishes at infinity. The constant and decaying parts of the equation give

$$a_0 b + \frac{1}{2} b^2 = 0, \quad (P+b)(D)f + \frac{1}{2} f^2 = 0.$$

The leading term of f at infinity forces $a_0 + b = 0$. It follows that $b = a_0 = 0$. Under the polynomial Laplace correspondence

$$\sum_k A_k \rho^k \mapsto \sum_k k! A_k z^{-k-1},$$

differentiation corresponds to multiplication by $-\rho$, and multiplication of the rational functions corresponds to continuous convolution. After a nonzero scalar normalization, the equation for f therefore gives $A \mid H_A$ for a monic A of degree $p-1$. By (R_{p-1}) , $A = \rho^{p-1}$. Thus $f = c_* z^{-p}$, and all the variables in (4.18) vanish. The leading equations have no nonzero common zero.

Step 4: the length. The same finite-cover Bézout argument as in Theorem 4.3, now with the weights (4.20), gives

$$\text{length } \mathcal{E}_p(g_2, g_3) = \frac{(p+1)(p+2) \cdots (2p-2)(2p)}{(p-2)! p} = 2 \binom{2p-2}{p-2}.$$

For $p=3$, this is $(4 \cdot 6)/(1 \cdot 3) = 8$. Empty products are one, so the calculation includes $p=2$. It proves this length at every (g_2, g_3) , including the singular parameters used below; at the cusp itself the whole length sits at the single solution where every unknown vanishes. It also makes the whole coefficient family finite over $\mathbb{C}[g_2, g_3]$: sufficiently large weighted monomials reduce to bounded weight using the leading equations, which are independent of g_2, g_3 . This finiteness is

what allows simplicity to be transferred from the nodal parameter value to nearby nonsingular parameters in the last step.

Step 5: the node. To prove generic simplicity, consider the nodal parameters

$$g_2 = \frac{1}{12}, \quad g_3 = -\frac{1}{216}. \quad (4.21)$$

In the logistic coordinate they are realized by

$$X = Q^2 - Q + \frac{1}{12}, \quad Y = (2Q - 1)Q(1 - Q), \quad DQ = Q(1 - Q).$$

The nodal coordinate ring is

$$\mathbb{C}[X, Y] = \{f \in \mathbb{C}[Q] : f(0) = f(1)\}.$$

Indeed the right side is $\mathbb{C} + Q(1 - Q)\mathbb{C}[Q]$; it is generated by X, Y , since, with $S = Q(1 - Q)$, $S = 1/12 - X$, $SQ = (Y + S)/2$, and $Q^2 = Q - S$. The pole filtration is the ordinary polynomial degree in Q . Consequently the nodal specialization of (4.18) has the unique form

$$V = \beta + v_A, \quad A \text{ monic of degree } p - 1, \quad A(0) = 0,$$

where β is the common endpoint. The coefficient change is invertible and affine over $\mathbb{Q}$.

The matching equations become, as equations of schemes,

$$A \in \mathcal{Z}_p, \quad P = P_A - \beta, \quad \beta(\beta - 2P_A(0)) = 0. \quad (4.22)$$

To see this directly, the constant equation is $P(0)\beta + \beta^2/2 = 0$. The remaining polynomial identity is $(P + \beta)(D)v_A + v_A^2/2 = 0$, equivalent by Proposition 4.1 to $A \mid S_A$ and $P + \beta = P_A$. The coefficient changes and monic division used here work over arbitrary $\mathbb{C}$ -algebras, so the correspondence preserves local multiplicities.

If $\mathcal{X}_p$ is reduced, Proposition 4.6 gives S_p simple points of $\mathcal{Z}_p$, with $P_A(0) \neq 0$ at every point. Each quadratic in (4.22) then has two distinct roots. The nodal fiber has $2S_p = M_p$ simple points. In the finite coefficient family, the image of the closed locus where the matching Jacobian vanishes is closed. Its complement contains (4.21), and hence meets the nonsingular-lattice locus in a nonempty open set. Every fiber in that open set is reduced, proving the theorem. $\square$

The factor two in M_p retains the two choices of constant equilibrium. The transformation

$$(P, v) \mapsto (P - 2P(0), v + 2P(0))$$

preserves the equation and the lattice. At the nodal fiber it exchanges the two roots in (4.22). On special nonsingular lattices those choices may coalesce.

4.5 Further orders

The prime-index argument excluded solutions at infinity by reducing the continuous convolution modulo ℓ . The same argument proves (4.10) whenever $2p - 1 = \ell^e$ is an odd prime power. For $\deg A = (\ell^e - 1)/2$, the normalized continuous convolution satisfies

$$\ell^e H_A(\rho) \equiv u\rho^{\ell^e} \pmod{\ell}, \quad u \neq 0.$$

Every bilinear coefficient other than the leading one has denominator valuation at most $e - 1$, whereas the leading coefficient has valuation $-e$. This follows either from the factorial formula or from

$$\frac{i!j!}{(i+j+1)!} = \sum_{r=0}^j \frac{(-1)^r \binom{j}{r}}{i+r+1}.$$

This argument controls only the leading system. It gives the length, but no simple-root lifting statement for the discrete matching system.

More generally, the exact arithmetic certificates supplied with the paper establish (R_d) for every $d \leq 72$, hence

$$\text{length } \mathcal{X}_p = \binom{2p-1}{p-1}, \quad \text{length } \mathcal{E}_p(g_2, g_3) = 2 \binom{2p-2}{p-2} \quad (2 \leq p \leq 73). \quad (4.23)$$

Together with the ancillary proof, they also cover infinite families of higher orders. The certificate covers all complex coefficient vectors in these degrees. The ancillary proof and the full certificate are described in Appendix A. Equation (4.23) is a statement about lengths only. At composite indices we establish simplicity here for $p = 5$ only, where $2p - 1 = 9$ is the first composite index: a separate exact calculation proves that all 126 rational-exponential points are simple; the input and its exact output are supplied in the ancillary material. Theorem 4.7 then gives 112 distinct elliptic pairs on a generic fixed lattice at that order.

Corollary 4.8 (An order-seven consequence). *Among all monic operators of order seven, there are exactly 1716 normalized rational-exponential pairs at unit rate, all simple. On every fixed nonsingular lattice the elliptic matching scheme has length 1584, and on a generic such lattice its 1584 points are distinct and simple.*

This follows from $2p - 1 = 13$. As throughout this section, the operators vary: the corollary counts pairs, not the waves of a single seventh-order equation.

5 Three classical equations

The three examples make both counts concrete and connect them to the classical literature. The first two use the full operator families of orders two and three, so their counts are the values N_2, M_2 and N_3, M_3 of the general formulas. The Kawahara equation restricts the order-four operator to an even polynomial; its counts must be computed within that restriction and differ from N_4 and M_4 . Table 1 summarizes the results. The normalizations are those of Section 4: for rational-exponential profiles the rate is one and the left endpoint is zero, reflection is retained, and for elliptic pairs both choices of constant equilibrium are retained.

5.1 KdV–Burgers and its KdV limit

The integrated travelling-wave equation for

$$u_t + uu_x + \nu u_{xx} + \delta u_{xxx} = 0, \quad \delta \neq 0,$$

has order $p = 2$. Its exact front solutions have been studied in several equivalent forms [22, 27, 43]. The long-wave reductions in Kudryashov et al. [32] also contain this balance of dissipation and dispersion.

Table 1: Counts of complex operator–profile pairs in the stated families. The elliptic column counts distinct pairs on a generic fixed nonsingular lattice; its value is the total multiplicity on every such lattice.

Equation	Monic operator $P(D)$	Rational-exponential	Elliptic
KdV–Burgers	$D^2 + a_1 D + a_0$	3	2
Kuramoto–Sivashinsky	$D^3 + a_2 D^2 + a_1 D + a_0$	10	8
Kawahara	$D^4 + b D^2 + a_0$	3	6

Here $A(\rho) = \rho + a$ has a single coefficient, and the matching system is one equation in one unknown. Direct polynomial division gives

$$\text{rem}_A S_A = -\frac{a(a-1)(a+1)}{6}, \quad P_A = \rho^2 + 5a\rho + a^2 - 6a - 1, \quad v_A = 12((a+1)Q - Q^2). \quad (5.1)$$

Thus there are exactly three simple pairs:

$$\begin{array}{c|c|c} a & P_A(\rho) & v_A \\ \hline -1 & (\rho-2)(\rho-3) & -12Q^2 \\ 0 & \rho^2 - 1 & 12Q(1-Q) = 3 \operatorname{sech}^2(z/2) \\ 1 & (\rho-1)(\rho+6) & 12Q(2-Q). \end{array} \quad (5.2)$$

The two fronts have nonzero Burgers coefficient, while the pulse belongs to the KdV limit $\nu = 0$. To return to the original equation with a general rate $\kappa \neq 0$, set

$$u(x, t) = u_* + \delta \kappa^2 v_A(\kappa(x - ct)), \quad \nu = \delta \kappa(5a), \quad u_* - c = \delta \kappa^2(a^2 - 6a - 1).$$

Differentiating the normalized equation verifies the original PDE. In particular, a nonzero ν fixes the front rate to $\kappa = \nu/(5a\delta)$, $a = \pm 1$, in agreement with the classical parameter restriction.

For a fixed lattice the elliptic candidate is $V = -12X + b$. Its residual is

$$-12a_1 Y - 12(a_0 + b)X + 6g_2 + a_0 b + \frac{1}{2}b^2.$$

Consequently the complete elliptic system is

$$a_1 = 0, \quad v = -12\wp - a_0, \quad a_0^2 = 12g_2. \quad (5.3)$$

There is no one-pole elliptic member with nonzero Burgers coefficient. For $g_2 \neq 0$ the fixed lattice gives two simple pairs. For $g_2 = 0$, $g_3 \neq 0$, it gives one pair of multiplicity two. This is $M_2 = 2$, including its exceptional specialization.

At the fixed operator $P = D^2 - 1$, equation (5.3) fixes $g_2 = 1/12$ and leaves g_3 free, subject to nonsingularity. These solutions lie on different energy levels, as Theorem 2.1 requires. At $g_3 = 0$, a translate by the imaginary half-period is

$$v(z) = 1 + \sqrt{3} \operatorname{cn}^2 \left(\frac{z}{\sqrt{2\sqrt{3}}} \middle| \frac{1}{2} \right). \quad (5.4)$$

The second argument is the Jacobi parameter [41, Chapters 22–23]. In the nodal limit $g_3 = -1/216$, the bounded translate becomes $3 \operatorname{sech}^2(z/2)$, the middle row of (5.2). The cusp $g_2 = g_3 = 0$ gives $P = D^2$, $v = -12z^{-2}$.

Since $2p - 1 = 3$ is prime, the one-pole counts at order two are unconditional. Global meromorphic completeness from Theorem 2.1 applies to the KdV subfamily $a_1 = 0$. For mixed P , the Fisher example in Section 6 explains why the same global conclusion cannot be imposed.

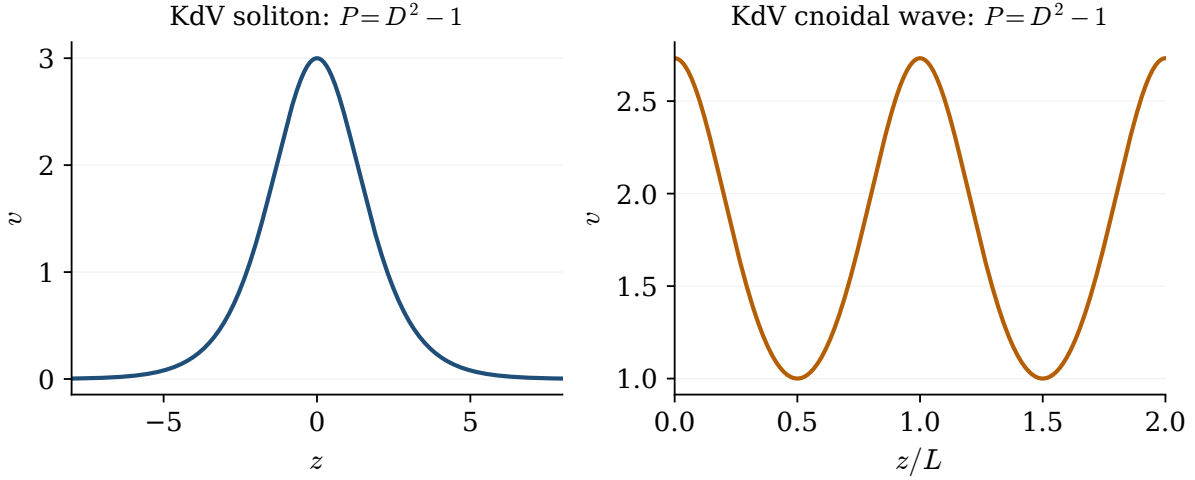


Figure 1: Two waves for the same normalized KdV operator $P = D^2 - 1$: the soliton in (5.2) and the cnoidal profile (5.4). Here $L = 2\sqrt{2\sqrt{3}} K(1/2)$ is the latter's real period.

5.2 Kuramoto–Sivashinsky: ten pairs and lattice collisions

Let $p = 3$, $s_3 = 60$, and $A(\rho) = \rho^2 + a\rho + b$. The profile and monic operator are

$$\frac{v_A}{60} = 2Q^3 - (a + 3)Q^2 + (a + b + 1)Q, \quad (5.5)$$

$$P_A(\rho) = \rho^3 + 4a\rho^2 + (a^2 + 19b)\rho - a^3 + 7ab - 5a - 30b. \quad (5.6)$$

The remainder of S_A is $(E_1\rho + E_0)/30$, where

$$E_0 = ab(a^2 - 7b + 5), \quad E_1 = a^4 - 8a^2b + 11b^2 + 10b - 1. \quad (5.7)$$

The system $E_0 = E_1 = 0$ is solved by cases on the factors of E_0 . If $b = 0$, then $a^4 = 1$. If $a = 0$, then $(b + 1)(11b - 1) = 0$. In the remaining case $a^2 = 7b - 5$, and the second equation becomes $4(b - 2)(b - 3) = 0$. This exhausts the system and gives Table 2. The Jacobian determinant is nonzero at every point, so all ten pairs are simple, as Theorem 4.4 predicts for $2p - 1 = 5$.

Table 2: All ten rational-exponential pairs at order three. The determinant is taken in the order (E_0, E_1) against (a, b) . Reality refers to the unit-rate logistic representatives.

a	b	Number	$\det \partial(E_0, E_1)/\partial(a, b)$	Real profiles
± 1	0	2	-24	Pulses
$\pm i$	0	2	-16	Not real
0	-1	1	144	Front
0	1/11	1	576/121	Front
± 3	2	2	-144	Fronts
± 4	3	2	384	Fronts

There are ten complex pairs, eight with real coefficients. All eight have smooth bounded representatives on the real axis: two pulses and six fronts. For example,

$$A = (\rho - 1)(\rho - 2), \quad v_A = 120Q^3, \quad P_A = (\rho - 3)(\rho - 4)(\rho - 5).$$

The choice $A = \rho(\rho + 1)$ gives the positive pulse $v_A = 120Q(1 - Q)^2$. These belong to the classical hyperbolic-function solution sector [24, 25, 37]. The complete parameter list is tabulated by Kudryashov and Zargaryan [31, Table 1] and Conte and Musette [4, Table 1]. Their tables contain six parameter cases because they identify the two signs of the dispersion coefficient: two cases have zero dispersion and four have nonzero dispersion, and restoring the sign of the latter gives the ten pairs of Table 2. Exhaustion of this classical list follows from Eremenko [18].

For comparison with the usual integrated KS equation, write

$$P = D^3 + BD^2 + \mu D + a_0, \quad w = v + a_0.$$

Then

$$w''' + Bw'' + \mu w' + \frac{1}{2}w^2 + K = 0, \quad K = -\frac{a_0^2}{2}. \quad (5.8)$$

An elliptic solution must have zero residue at its pole. The Laurent residue is $60\mu/19 - 15B^2/76$, and its vanishing gives $B^2 = 16\mu$. The resulting elliptic formula is

$$w = -60\wp' - 15B\wp - \frac{B^3}{64}, \quad g_2 = \frac{B^4}{3072}, \quad g_3 = \frac{K + 13B^6/4096}{1080}. \quad (5.9)$$

This is the classical solution of Kudryashov [25]; necessity of its elliptic parameter restriction is proved by Hone [20], and the classification of all meromorphic KS solutions is completed by Eremenko [18]. With leading coefficient ν , the residue condition is $B^2 = 16\nu\mu$.

The classical formula gives the lattice in terms of the operator. For enumeration we read it the other way: fix g_2, g_3 and solve for B, a_0 . The entire elliptic matching scheme is

$$\begin{aligned} P &= D^3 + BD^2 + \frac{B^2}{16}D + a_0, \\ v &= -60\wp' - 15B\wp - \frac{B^3}{64} - a_0, \\ B^4 &= 3072g_2, \quad a_0^2 = \frac{13B^6}{2048} - 2160g_3. \end{aligned} \quad (5.10)$$

The coefficients eliminated along the way are determined linearly with nonzero rational pivots, so the last two equations carry the full matching system, multiplicities included. Their quotient algebra has basis $B^i a_0^j$, $0 \leq i < 4$, $0 \leq j < 2$, and length eight, which is M_3 .

Generically these are eight distinct pairs; the exceptional cases are found as follows. Let

$$j(\Lambda) = \frac{1728g_2^3}{g_2^3 - 27g_3^2}.$$

The Jacobian of the last two equations in (5.10) is a nonzero constant times $B^3 a_0$. If $B = 0$, then $g_2 = 0$, hence $j = 0$. If $a_0 = 0$ and the lattice is nonsingular, elimination gives

$$675g_3^2 = 169g_2^3, \quad j = -300.$$

These conditions are sufficient as well. Thus:

$j(\Lambda)$	distinct pairs	local multiplicities
$j \notin \{0, -300\}$	8	1, 1, 1, 1, 1, 1, 1, 1
-300	6	2, 2, 1, 1, 1, 1
0	2	4, 4.

(5.11)

For $j = -300$, all four B -roots are simple, and precisely two give $a_0 = 0$, producing the two double points. For $j = 0$, the equations reduce to $B^4 = 0$, $a_0^2 = -2160g_3$, producing two points of length four. For instance, $g_2 = 1/12$, $g_3 = 13/1080$ gives the double points $(B, a_0) = (4, 0), (-4, 0)$. This calculation exhibits explicitly the exceptional set in Theorem 4.7.

The exponential and elliptic counts refer to varying operators. No operator in Table 2 also has a nonsingular elliptic solution: the six unequal-endpoint pairs violate the residue condition, while the four equal-endpoint pairs give singular invariants in (5.9). This agrees with uniqueness up to translation for fixed P . For $P = D^3$, the rational member is $v = 120z^{-3}$.

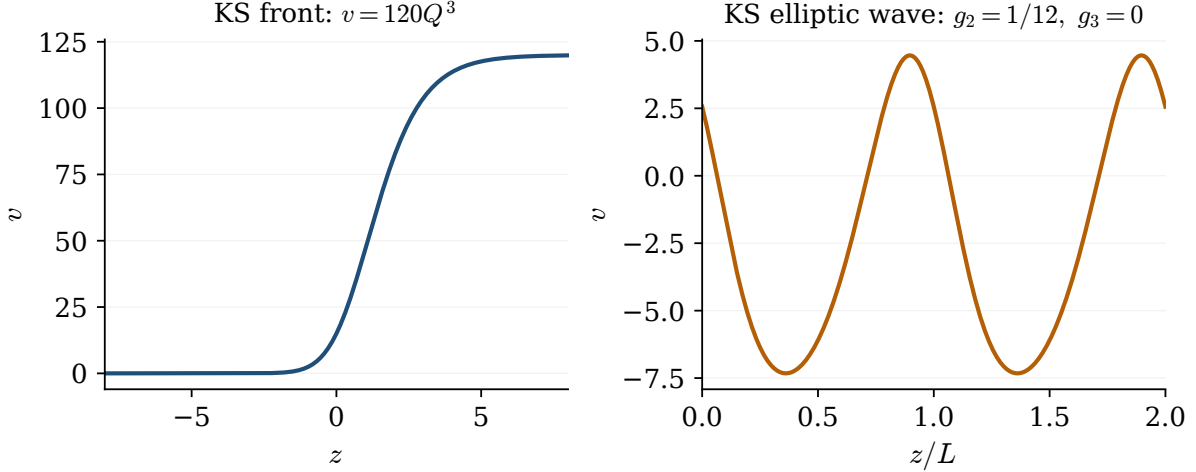


Figure 2: A KS front $v = 120Q^3$ and a bounded elliptic wave. For the latter, $g_2 = 1/12$, $g_3 = 0$, $P = D^3 + 4D^2 + D + \sqrt{26}$, and $v = -60\wp' - 60\wp - 1 - \sqrt{26}$, translated by the imaginary half-period; $L = 2\sqrt{2\sqrt{3}} K(1/2)$ is its real period. The operators in the two panels are different.

There is also a direct connection with the vessel model of Kudryashov et al. [32, Eq. (4.12)]. Its quadratic long-wave equation has coefficients $-\lambda/(2\alpha)$, $(\gamma - \beta)/(2\alpha)$ and $\delta/(2\alpha)$ at the second, third and fourth spatial derivatives. When its quadratic advection coefficient and $\alpha\delta$ are nonzero, amplitude normalization puts it in the same travelling-wave class. The elliptic condition becomes

$$(\gamma - \beta)^2 + 16\delta\lambda = 0.$$

Real material coefficients with $\delta\lambda > 0$ therefore exclude the elliptic sector. The rational-exponential sector is tested by (5.7), with rate and coefficient scales restored.

5.3 Kawahara: an even coefficient restriction

For the Kawahara equation [14, 23], the integrated, shifted and monic travelling-wave operator is

$$P = D^4 + bD^2 + a_0. \quad (5.12)$$

Meromorphic completeness for this equation, using its first integral, is already established by Demina and Kudryashov [14]; see also Ye et al. [44]. Their first integral is the order-four instance of (2.7). Theorem 2.1 places that argument in the uniform operator setting, with the energy retained when fixing a Laurent expansion.

The even restriction changes the rational-exponential count, and the reason is a symmetry. An even operator gives a profile even about its pole: reflection preserves both the equation and its energy, so Laurent uniqueness applies. In the logistic coordinate this is $v(Q) = v(1-Q)$. Since $v(0) = 0$, both endpoints are zero; the normalized polynomial must have the form $A = \rho^3 + c\rho$. Its quotient and remainder are

$$\begin{aligned} P_A &= \rho^4 + 13c\rho^2 + \frac{31c^2 - 70c + 7}{3}, \\ v_A &= 280(1+c)Q(1-Q) - 1680Q^2(1-Q)^2, \\ \text{rem}_A S_A &= -\frac{\rho}{420}(c+1)(31c^2 - 31c + 10). \end{aligned} \tag{5.13}$$

The three distinct roots are

$$c = -1, \quad c = \frac{1}{2} \pm \frac{3i}{2\sqrt{31}}.$$

They give the three simple rational-exponential pairs in this coefficient family. These are the three simply periodic parameter choices of Demina and Kudryashov [14, Section 3], in the present normalization: the parameter in their Eq. (31) equals $1/c$ here. At $c = -1$,

$$v = -105 \operatorname{sech}^4(z/2), \quad P = (D^2 - 4)(D^2 - 9). \tag{5.14}$$

This is the familiar solitary-wave branch of the Kawahara equation; the other two pairs are complex at unit real rate. The count three is for the even family (5.12). The unrestricted order-four count in Theorem 4.4 is 35.

All elliptic members of (5.12) have the form

$$v = -1680\wp^2 - \frac{280b}{13}\wp + 168g_2 + \frac{31b^2}{507} - a_0. \tag{5.15}$$

Substitution gives exactly two conditions:

$$F(b; g_2, g_3) := 31b^3 - 42588bg_2 - 4745520g_3 = 0, \tag{5.16}$$

$$a_0^2 = 23184g_2^2 + \frac{7980}{169}b^2g_2 + \frac{101520}{13}bg_3. \tag{5.17}$$

These are the Weierstrass profiles and parameter relations obtained from Laurent series by Demina and Kudryashov [14, Section 3], expressed in the normalization (1.1). In their notation, $b = -\beta$, $a_0^2 = C_0^2 + 12C_1$; the profile change is $v = -6(w - w_*)$, where w_* is a constant equilibrium and $a_0 = C_0 - 6w_*$. The same two equations give the fixed-lattice count directly: the first is cubic in b , and the second is monic quadratic in a_0 . Their quotient algebra has dimension six for every fixed lattice. At $(g_2, g_3) = (1, 0)$, the cubic has three distinct roots and the right side of (5.17) is nonzero at all of them. Thus the six points are simple on a nonempty open set of lattices. This is the count within the even family; the full order-four elliptic matching scheme has length 30. Elliptic formulas with nonzero odd derivative coefficients in the full fifth-order evolution equation are given by Kudryashov and Sinelshchikov [35].

For a fixed-operator illustration, set $b = 0$. Equations (5.16)–(5.17) then give

$$P = D^4 + a_0, \quad g_3 = 0, \quad g_2 = \pm \frac{a_0}{12\sqrt{161}}. \tag{5.18}$$

If $a_0 \neq 0$, both lattices are nonsingular. The corresponding profiles

$$v = -1680\wp^2 + 168g_2 - a_0$$

have marked coefficients $h = [z^8]v = -21g_2^3/130$, and hence different energies. There is no nonconstant rational or rational-exponential solution for this fixed operator: at a singular cubic $g_2 = q^2/12$, $g_3 = -q^3/216$, the first equation forces $q = 0$, and the second forces $a_0 = 0$. When $a_0 = 0$, only the rational profile $v = -1680z^{-4}$ remains. These statements exhaust the nonconstant globally meromorphic solutions for $P = D^4 + a_0$, up to translation.

At $a_0 = 12\sqrt{161}$, $g_2 = 1$, $g_3 = 0$, a translate by the imaginary half-period is the real bounded wave

$$v(z) = -1680 \left(-\frac{1}{2} + \frac{1}{2} \operatorname{sn}^2(z \mid 1/2) \right)^2 + 168 - 12\sqrt{161}. \quad (5.19)$$

Its real period is $2K(1/2)$. Figure 3 compares it with (5.14). The energy parameter is what permits distinct lattices for a fixed even operator, while preserving uniqueness at fixed marked data.

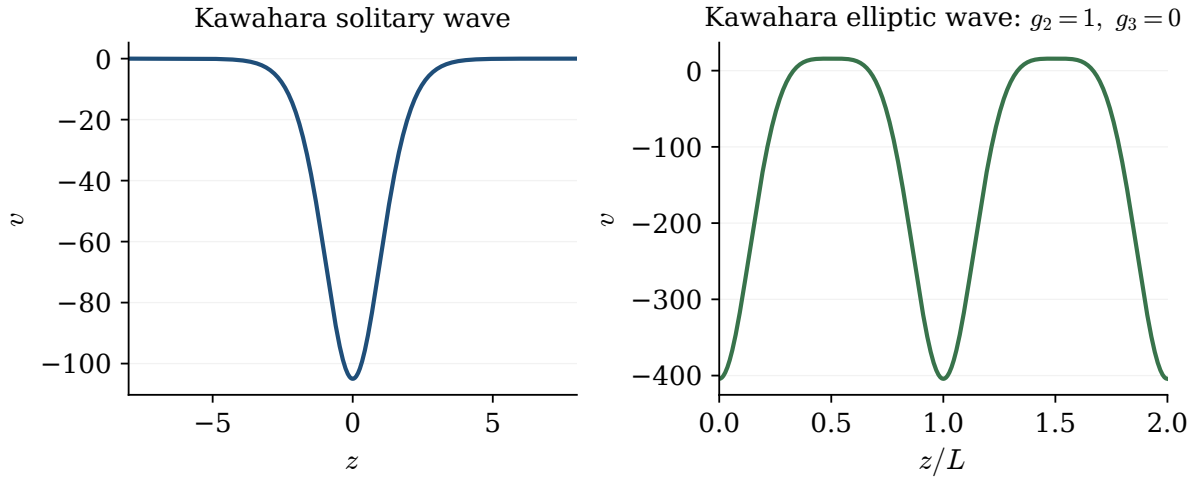


Figure 3: A solitary wave and a bounded elliptic wave in the Kawahara family. The left panel is (5.14); the right panel is (5.19) for $P = D^4 + 12\sqrt{161}$. The latter's horizontal coordinate is divided by its real period.

6 Open questions and the boundary of the classification

Two questions remain open in the enumeration, and one limitation of the classification deserves a precise statement.

Rigidity. The first question is the continuous-convolution rigidity assertion

$$0 \neq A \in \mathbb{C}[\rho], \quad A \mid \int_0^\rho A(t)A(\rho-t) dt \implies A = c\rho^{\deg A}.$$

Through the polynomial Laplace correspondence used in the proof of Theorem 4.7, it says that, after fixing the pole at zero, any rational one-pole solution of (1.1) vanishing at infinity would have to be c_*z^{-p} , with operator $P = D^p$. Algebraic and p -adic arguments prove it in the range $\deg A \leq 72$, corresponding to $2 \leq p \leq 73$, and in infinite arithmetic families of higher degrees. Rigidity in arbitrary degree remains a conjecture. By Theorems 4.3 and 4.7, it would establish both enumeration formulas with algebraic multiplicities at every order.

Simplicity. The second question is whether every point of $\mathcal{X}_p$, the rational-exponential matching system, is simple. The elliptic simplicity assertion is generic in the lattice and is derived from this one in Theorem 4.7. At a solution $S_A = AC_A$, the tangent map of the system is

$$J_A(h) = \text{rem}_A(2(A \star h) - C_A h), \quad \deg h < d. \quad (6.1)$$

The question is whether this map always has zero kernel. The prime-index argument proves this after reduction modulo ℓ , where the map becomes multiplication by a polynomial coprime to $\overline{A}$; a uniform replacement is missing at composite indices. Laurent uniqueness concerns a fixed operator, whereas (6.1) allows the operator to vary. It therefore does not by itself prove simplicity of $\mathcal{X}_p$.

Conjecture 6.1. Continuous-convolution rigidity holds in every degree, and the matching Jacobian (6.1) is invertible at every point of $\mathcal{X}_p$. Consequently $\#\mathcal{X}_p(\mathbb{C}) = \text{length } \mathcal{X}_p = \binom{2p-1}{p-1}$ for every $p \geq 2$.

The conjecture would also imply M_p simple elliptic pairs on a generic fixed lattice at every order. Simplicity on every lattice is false already at order two; (5.11) describes the full exceptional set at order three.

Mixed even operators. The even-operator hypothesis in Theorem 2.1 is a separate issue, and it cannot simply be dropped. The Fisher equation supplies a known illustration of the limitation: special-speed solutions were obtained by Ablowitz and Zeppetella [1], and exact solitary representations were also studied by Kudryashov [26]. Its elliptic-exponential family is also given in Kudryashov [28], Conte et al. [5, Lemma 4.4], and McCue et al. [38]. In our normalization it is

$$v(z) = -12\lambda^2 e^{2\lambda z} \wp(e^{\lambda z} - 1; 0, g_3), \quad P(D) = D^2 - 5\lambda D + 6\lambda^2, \quad \lambda \neq 0.$$

For generic $g_3 \neq 0$, this is globally meromorphic and lies outside $\mathcal{W}$. Further examples of meromorphic solutions outside $\mathcal{W}$ are discussed by Ng and Wu [40, Remark 1.11]. Substitution follows from $(D^2 - 5\lambda D + 6\lambda^2)[t^2 U(t)] = \lambda^2 t^4 U''(t)$, $t = e^{\lambda z}$. Thus an unrestricted mixed even-order theorem requires a broader global conclusion.

The energy argument does have a partial analogue for some mixed operators, which admit a polynomial Darboux identity $(D + 3\mu)\mathcal{J} = (v' + \mu v)E_P(v)$. For identities with $\mathcal{J} = v^3/6 +$ quadratic jet terms, the zero level has the same one-pole argument, while nonzero levels satisfy the nonautonomous relation $\mathcal{J} = Ce^{-3\mu z}$. A concrete higher-order test is $P = (D + 2)(D + 3)((D + 3)^2 + 1)$, with marked coefficient $[z^8]v = 0$: the local Laurent condition is compatible and the Darboux constant is nonzero, as checked in the ancillary material. This also shows that evenness is sufficient, but not necessary, for local resonance compatibility. It is not covered by the even-operator or zero-level argument. Methods for constructing elliptic and rational-exponential solutions without the finiteness property are developed by Demina [13]. The global classification of solutions on nonzero Darboux levels of general mixed even operators is not addressed by the present argument.

A Exact verification and ancillary material

This appendix records one lemma used in the proof of Theorem 4.4 and describes the computer checks that accompany the paper.

The lemma transfers the absence of solutions at infinity from characteristic ℓ to characteristic zero. Let homogeneous polynomials $F_1, \dots, F_m$, with coefficients in $\mathbb{Z}_{(\ell)}$, have no nonzero common zero over $\overline{\mathbb{F}}_\ell$. Then they have no nonzero common zero over $\mathbb{C}$. Indeed a complex projective zero yields a zero over $\overline{\mathbb{Q}}$ after fixing one nonzero coordinate. Choose a place above ℓ , extend it if necessary, and scale all coordinates by the inverse of a coordinate of smallest valuation. The coordinates are integral and one is a unit. Their reductions give a nonzero common zero over $\overline{\mathbb{F}}_\ell$, contradicting the hypothesis. The argument needs the algebraic closure of the residue field; a search only over $\mathbb{F}_\ell$ would not suffice.

The ancillary material contains three kinds of checks.

1. `verify_manuscript.py` reconstructs the discrete and continuous convolutions, the low-order Laurent coefficients, and the Weierstrass differential algebra. It also checks the leading-term comparison used in Proposition 4.2. It checks the displayed order-two, order-three and order-four formulas by exact substitution, including the correspondence with the KS tables of Conte and Musette [4], Kudryashov and Zargaryan [31], the KS collision multiplicities, the Kawahara coefficient restriction and its literature normalization, the nodal correspondence, and the Fisher identity. These finite substitutions check the displayed formulas; the uniform proofs are the arguments in the text.
2. `p5_count.sing` is a self-contained rational SINGULAR calculation for the full order-five matching ideal. Its quotient dimension is 126. Adjoining the matching Jacobian determinant gives the unit ideal, so all 126 points are simple. The input is reconstructed from the original polynomial sum by the Python checker; the saved transcript states the two exact outputs.
3. `rigidity/` supplies the proof and exact certificate for (R_d) , $d \leq 72$. For $2d + 1 = \ell^e + g$, $g \in \{0, 2, 4, 6\}$, the proof reduces the normalized continuous convolution to a residual system of degree at most g . The certificate retains every residual degree, with integer ideal-membership identities and certified prime factors. The replay reconstructs the inputs, multiplies out every identity, and checks all 72 arithmetic choices. It does not infer characteristic-zero nonexistence from a list of residue-field points.

A Lean 4 formalization accompanies the paper at <https://github.com/migita/binomial-count-meromorphic-waves>, directory `lean/`, with Lean and Mathlib pinned to version 4.33.1. It proves (R_d) for $2d + 1$ prime (`rigidity_prime_index`) and the count and simplicity statements of Theorem 4.4, both for the matching system (4.6) (`primeIndexCountAndSimplicity`) and for the normalized pairs (P, v) as solutions of (1.1) in the field of rational functions of e^z (`primeIndexRationalExponentialCountAndSimplicity`), together with the correspondence between the two formulations for every $p \geq 2$. The discrete convolution is defined there by the Bernoulli-polynomial power sums, and the remainder equations and their Jacobian are those of Section 4. The proof route is described in Remark 4.5. The files `EnumerationCheck.lean` and `Check.lean` verify that the proofs have exactly the stated propositions as their types and that the theorems depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`; the project contains no `sorry`, added axiom or `native_decide`.

The scripts use exact rational or finite-field arithmetic. The figure program evaluates the already verified formulas. Instructions, dependency versions and checksums are included with the source package. The prime-index theorem does not depend on the computer calculations.

Tool and computational resource disclosure

This paper was written with substantial use of generative AI, which is disclosed here in the form recommended by the Leiden Declaration on Artificial Intelligence and Mathematics¹ and required by arXiv. No automated system is an author; credit and responsibility for the results, and for the attribution of prior work, rest with the author.

- **gpt-6-astra** (OpenAI), August–September 2026. The main mathematical ideas of the paper, the proofs of Theorems 2.1, 4.3, 4.4 and 4.7, the rigidity condition (R_d) and its certificate, the exact verification scripts, the SINGULAR input, the first complete draft of the text, and the Lean 4 formalization of Remark 4.5 were produced by this model in dialogue with the author.
- OpenAI Codex v0.153.4, September 2026. Independent mathematical audit of the final revision, integration of the edits, and rebuilding and validation of the source package.
- Claude Code (Anthropic, Claude Fable 5.1), September 2026. The literature search behind the attributions in Section 1, the readability revision of all sections, and the leading-coefficient argument of Proposition 4.2, which Codex then checked independently. It also audited the Lean formalization independently: the formal statements against the manuscript, a rebuild from source, and the axiom audit.

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